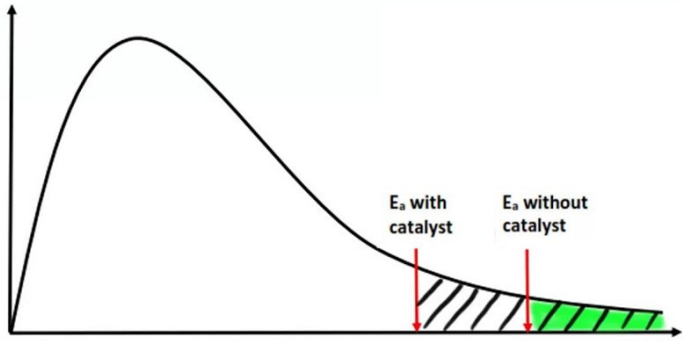


Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iv)		award (1) each for any two of following • NO/NO₂/CO are toxic • CO₂ contributes to greenhouse effect / global warming • NO₂/NO_x contribute to acid rain		2		2		
		(v)		exhaust (1) award (1) for any of following • heterogenous • a transition metal / a transition metal compound • palladium / platinum / rhodium	2			2		
	(c)			 shape of curve (1) lower activation energy with catalyst / shown on graph (1) more collisions / molecules have energy greater than activation energy (1)	3			3		
				Question 8 total	7	2	6	15	4	2